

Lecture 09: The Linear Model II

Nonlinear Transformations and Logistic Regression

Caltech CS156: Learning from Data

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1 Taxonomy of Linear Models

Linear models process input $\mathbf{x}$ via a linear signal $s = \mathbf{w}^T \mathbf{x}$. The output transformation defines the specific model type:

1. **Linear Classification:** $h(\mathbf{x}) = \text{sgn}(s) \in \{-1, +1\}$ (Hard threshold).
2. **Linear Regression:** $h(\mathbf{x}) = s \in \mathbb{R}$ (Identity).
3. **Logistic Regression:** $h(\mathbf{x}) = \theta(s) \in [0, 1]$ (Soft threshold / Probability).

2 Nonlinear Transformations $\mathbf{x} \rightarrow \mathbf{z}$

To fit non-linear decision boundaries, we map inputs $\mathbf{x} \in \mathbb{R}^d$ to a feature space $\mathbf{z} = \Phi(\mathbf{x}) \in \mathbb{R}^{\tilde{d}}$. Because the model remains linear in $\mathbf{w}$ within the $\mathcal{Z}$ -space, all linear algorithms apply directly. However, the VC dimension increases from $d + 1$ to $\tilde{d} + 1$.

Data Snooping Hazard

Choosing the feature transformation Φ after visually inspecting the dataset $\mathcal{D}$ constitutes **Data Snooping**. Doing so invalidates the VC bound warranty, leading to severe overfitting.

3 Logistic Regression Formulation

Logistic Regression estimates the conditional probability $P(y = +1 | \mathbf{x})$ using the **logistic sigmoid function**:

$$\theta(s) = \frac{e^s}{1 + e^s} = \frac{1}{1 + e^{-s}} \quad (1)$$

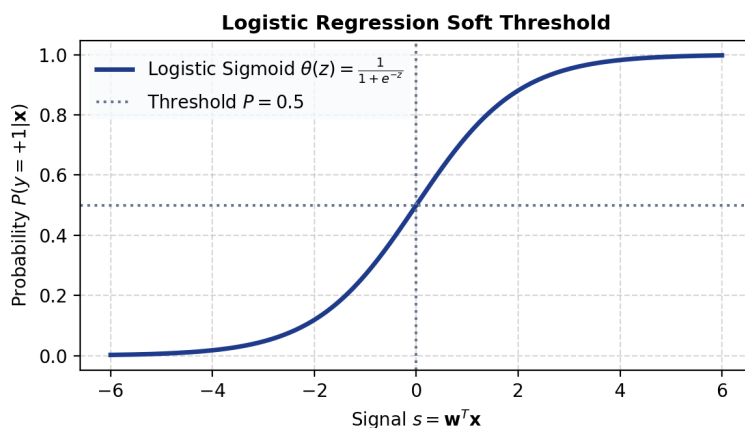


Figure 1: The logistic sigmoid function mapping signal $s = \mathbf{w}^T \mathbf{x}$ to probability $P \in (0, 1)$.

3.1 Maximum Likelihood Estimation & Cross-Entropy Loss

Assuming binary targets $y_i \in \{-1, +1\}$, the probability of observing label y_i given $\mathbf{x}_i$ is:

$$P(y_i | \mathbf{x}_i) = \theta(y_i \mathbf{w}^T \mathbf{x}_i) \quad (2)$$

Assuming independent observations, the likelihood of the entire dataset $\mathcal{D}$ is:

$$L(\mathbf{w}) = \prod_{i=1}^N P(y_i | \mathbf{x}_i) = \prod_{i=1}^N \theta(y_i \mathbf{w}^T \mathbf{x}_i) \quad (3)$$

Maximizing likelihood $L(\mathbf{w})$ is equivalent to minimizing the negative log-likelihood divided by N , defining the **Cross-Entropy Error** $E_{\text{in}}(\mathbf{w})$:

Cross-Entropy Loss (Log Loss)

$$E_{\text{in}}(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N \ln \left(1 + e^{-y_i \mathbf{w}^T \mathbf{x}_i} \right) \quad (4)$$

4 Gradient Descent Optimization

$E_{\text{in}}(\mathbf{w})$ is a smooth, strictly convex function, guaranteeing a unique global minimum. The gradient vector is:

$$\nabla E_{\text{in}}(\mathbf{w}) = -\frac{1}{N} \sum_{i=1}^N \frac{y_i \mathbf{x}_i}{1 + e^{y_i \mathbf{w}^T \mathbf{x}_i}} \quad (5)$$

The parameter update rule at step t with learning rate η is:

$$\mathbf{w}(t+1) = \mathbf{w}(t) - \eta \nabla E_{\text{in}}(\mathbf{w}(t)) \quad (6)$$